

Ashley N. Prow-Fleischer

last update: Aug. 26

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Education & Training

2025 - Now Postdoctoral Scholar Stanford University | [Paleobiology Group](#) | PI. Dr: Jonathan Payne

2021 - 2025 Ph.D. Syracuse University | Paleoclimatology | PI: Dr. Zunli Lu

2014 - 2017 B.S. Florida State University | Environmental Chemistry

2014 - 2017 B.S. Florida State University | Biomathematics

Publications

Peer-Review Journals

Prow-Fleischer, A., Lu Z., Penn, J.L., Deutsch, C., Todes, J.P., Blättler, C.L., Day, J., Mucciarone D., Pohl, A., Ridgwell, A., Al Aswad J. A., Guthrie, R., Jenkyns, H.C., Robinson, S.A., Yanan Shen, Y., Razonale, D., He, R., Elrick, M., Payne, J. L. (**in submission**). Temperature-dependent physiological traits and biogeography link Late Devonian Kellwasser Extinction to climate cooling.

Bazzi, M., Prow-Fleischer, A., Siversson, M., Al Aswad, J.A., Ebersole, J.A., Payne, J.L. (**in review**). Broad-scale biogeographic stability despite mass extinction in sharks across the K/Pg boundary. *Proc. R. Soc. B*

He, R., Pohl, A., Zhang, X., Chang, C., Prow-Fleischer, A. Payne, J.L., Xiao, S., Ridgwell, A., Lu, Z. (2026) A reversed latitudinal ocean oxygen gradient in the Proterozoic Eon. *Nat. Geosci.* **19**, 325–330. <https://doi.org/10.1038/s41561-025-01896-w>

Prow-Fleischer, A.N., & Lu, Z. (2025). A laser ablation ICP-MS protocol for high-resolution I/Ca profiling in corals. *Rapid Comm. Mass Spec.* <https://doi.org/10.1002/rcm.10002>

Prow-Fleischer, A.N., Lu, Z., Blättler, C.L., He, T., Yang, Z., Singh, P., Kemeny, P.C., Todes, J.P., Pohl, A., Bhattacharya, T., van de Schootbrugge, B., Wignall, P.B., Payne, J.L. (2025). Calcium isotopes support spatial redox gradients on the Tethys European margin across the Triassic-Jurassic boundary. *Chemical Geology*. <https://doi.org/10.1016/j.chemgeo.2024.122530>

Prow-Fleischer, A.N., Lu, Z., Meehan, K.C., Yang, Z., Ivany, L.C., Payne, J.L., (2024) Body-size reductions in dactyloconarid tentaculitoids during Late Devonian warming. *Geosphere*. <https://doi.org/10.1130/GES02759.1>

- He, R., Pohl, A., **Prow, A.N.**, Jiang, G., Huan C. C., Saltzman, M., Ridgwell, A., Lu, Z. (2024). The dynamic ocean redox evolution during the Late Cambrian SPICE event: evidence from the I/Ca proxy. *Global and Planetary Change*. <https://doi.org/10.1016/j.gloplacha.2024.104354>
- Lu, Z., Rickaby, R.E.M., Payne, J.L., **Prow, A.N.** (2024). Phanerozoic Coevolution of O₂-CO₂ and Ocean Habitability. *National Science Review*. <https://doi.org/10.1093/nsr/nwae099>
- Prow, A.N.**, Lu, Z., Frappier, A., Weisbeck, L., Underwood, C. (2023). Extraction of calcareous Dacryoconarid microfossils from limestones and mudrocks by surfactants paired with freeze-thaw processing. *Marine Micropaleontology*. <https://doi.org/10.1016/j.marmicro.2023.102216>
- Lu, Z., Thomas, E., Rickaby, R.E.M., Lu, W., and **Prow, A.N.** (2023). Commentary: Planktic foraminifera iodine/calcium ratios from plankton tows: *Frontiers in Marine Science*, <https://doi.org/10.3389/fmars.2023.1221835>

Grants and Fellowships (* competitive | ** collaborative)

2026	National Science Foundation ACCESS Explorer's Grant (EES260080 & CIS260896) 100 core hours**
2025	The Micropaleontology Society Angelina Messina Grant \$600
2024 - 2025	Research Excellence Doctoral Funding Graduate Fellowship (Syracuse University)*
2024	Geological Society of America (GSA) On-to-the-Future Mentor Program Travel \$600
2024	GSA/Chevron Field Trip Grant \$290
2024	William B. & Dorothy Heroy Research Grant \$2200
2023	Syracuse University Graduate Student Organization Travel Grant \$600
2023	Travel Grant Pre-GSA workshop, EarthCube Geochronology \$800
2023	Summer Research Grant, Merriam Endowment, Syracuse University Internal \$2000

Presentations

Invited Talks

April 2027 Environmental Fridays Season 12 | TBD

Feb 2026 Rochester Academy of Science Paleontology Division | “Linking anoxia, climate cooling, and marine extinction selectivity across the Kellwasser events”

Conference Talks

July 2025 Prow-Fleischer, A.N., Lu, Z., Blättler, C., Ridgwell, A., Payne, J., Penn, J., Al Aswad, J., Pohl, A. “Linking anoxia, climate cooling, and marine extinction selectivity across the Kellwasser events [using proxy-constrained Earth-system modelling approach]”. Goldschmidt 2025, Prague, Czech Republic

Commented [AR1]: I prefer the original ‘... modelling approach’
Unsure what ‘selection’ means in this context.

April 2024 Prow, A.N., Lu, Z., Blättler, C.L., He, T., Yang, Z., Singh, P., Kemeny, P.C., Todes, J.P., Pohl, A., Bhattacharya, T., van de Schootbrugge, B., Wignall, P.B., Todaro, S. Payne, J.L. “Calcium isotopes validate spatial redox gradients on the Tethys European margin across the Triassic-Jurassic boundary.” NE Geobiology Symposium, New Haven, CT.

Commented [AP2R1]: I agree

Dec.2023 Prow, A.N., Lu, Z., Blättler, C.L., He, T., Yang, Z., Singh, P., Kemeny, P.C., Todes, J.P., Pohl, A., Bhattacharya, T., van de Schootbrugge, B., Wignall, P.B., Payne, J.L. “Temporal and spatial dynamics of paleo-redox conditions across the Triassic-Jurassic boundary.” American Geophysical Union Annual Meeting. San Francisco, CA

Oct. 2023 Prow, A.N., Lu, Z., Meehan, K.C., Yang, Z., Ivany, L.C., Payne, J.L., “Dacryoconarid Body Size Trends across the Late Devonian Punctata Event: Insights into their Evolutionary Responses to Paleoenvironmental Shifts.” Geological Society of America Connects , Pittsburg, PA

Aug. 2023 Prow, A.N., Lu, Z., Meehan, K.C., Yang, Z., Ivany, L.C., Payne, J.L., “Exploring Dacryoconarid Body Size Trends across the Late Devonian Punctata Event: Insights into their Evolutionary Responses to Paleoenvironmental Shifts.” Paleontological Research Institute Summer Symposium

Posters

Sept. 2024 Prow-Fleischer, A.N. & Lu, Z. “A laser ablation ICP-MS protocol for high-resolution I/Ca profiling in corals.” GSA Connects Annual Conference, Anaheim, CA

Sept. 2023 Prow, A.N., Lu, Z., Meehan, K.C., Yang, Z., Ivany, L.C., Payne, J.L., “Dacryoconarid body size and diversity trends across the *punctata* carbon isotope excursion, Appalachian Basin” International Sub-Commission for Devonian Stratigraphy conference, Geneseo NY.

Student Presentations

- Dec. 2026 Mu, J.K., Prow-Fleischer, A., Payne, J.L. "Applying Distance Based Sensitivity Analysis to Quantify Variable Uncertainty in a Carbon-Calcium Isotope Mass Balance Model"
AGU Annual Meeting
- Dec. 2026 Kwak, Z., Prow-Fleischer, A., Payne, J.L. "Phosphorus Cycle is an Important Component in Reconciling Ca-C Isotope Records Across Late Devonian Upper Kellwasser Event"
AGU Annual Meeting

Mentori

Teaching Experience

Instructor

- Spring 2026 Topics in Geobiology | Stanford University
- Summer 2024 Instructor: Climate Change: Past, Present and Future | Syracuse University

Graduate Teaching Assistant

Introduction to Oceanography, Climate Change Past and Present

Professional Teaching

- 2018 - 2021 Florida State University Department of Chemistry
- Courses instructed: Introduction to Analytical Chemistry, Advanced (Instrumental)
Analytical Chemistry, Physical Chemistry

Professional Experience

- 2018 – 2021 Undergraduate Chemistry Laboratory Teaching Specialist | Florida State University
- 2017 - 2018 Lead Lab Technician | Surterra Wellness

Professional Service

Professional Societies

- April 2025 - Geological Society of America Student Advisor Council member (2 year term)

Sept 2024 Organizer of Geological Society America '24 (Anaheim) Topical Sessions Part I and II
"Piecing together the extinction puzzle: Integrated approaches to quantifying organism-
environment interactions."

Ad-hoc Reviewer

Marine Micropaleontology, Science Advances

Departmental Committees

2024 - 2025 Syracuse University Chief Diversity Officer, Department Graduate Student Organization
(GeoGo)

2022 - 2023 Student Affairs Coordinator, Department Graduate Student Organization (GeoGo)

2022 - Member Diversity, Equity, and Inclusion Committee, Syracuse University Department of
Earth and Environmental Sciences

University Organizations

2022 - 2024 Member of Women in Science and Engineering Future Professionals Program (WISE-
FFP) at Syracuse University (Certificate conferred 2024)

Awards and Honors

2025 Syracuse University College of Arts and Sciences Doctoral Prize

2024 GSA Foundation Penrose Circle Heroy Student Award

2024 Marjorie Hooker Outstanding Thesis proposal

2024 Honorable Mention Syracuse University Dean's Award

2023/24 Publication Award, Syracuse University Department of Earth & Environmental Sciences

2013 Women's History Month Excellence in STEM award, Tallahassee Community College

Community Engagement

2022-2025 Board Member for Friends of Clark Reservation State Park

Certifications

2026 Certificate in High Performance Research Computing | Issued by Stanford University